第11章 光流计算

计算机科学与技术学院

華東師範大學
EAST CHINA NORMAL UNIVERSITY

本次课程内容

1. 什么是光流
2. 光流估计的用途
3. 运动场与光流关系
4. 光流的估计方法
5. 密集光流与稀疏光流
6. 智能光流估计技术及应用

1. 什么是光流（optical flow）

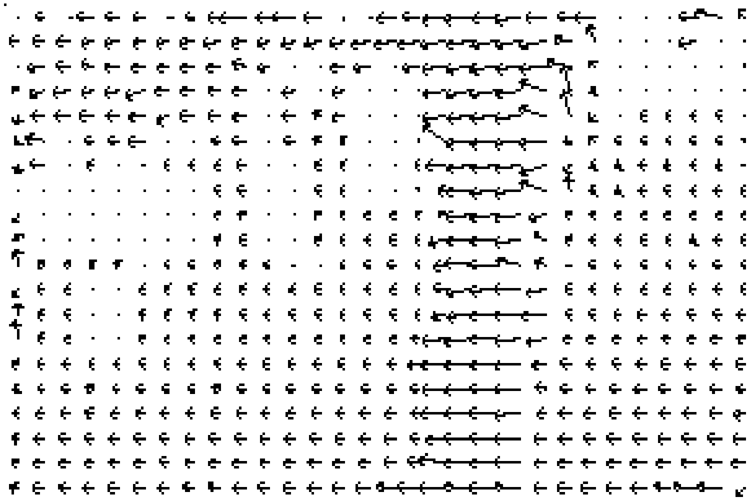
◆ 光流:是空间运动物体在成像平面上像素运动的瞬时速度

前一帧

后一帧

from S. Seitz, R. Szeliski, M. Pollefeys

光流实例



from S. Seitz, R. Szeliski, M. Pollefeys

光流实例

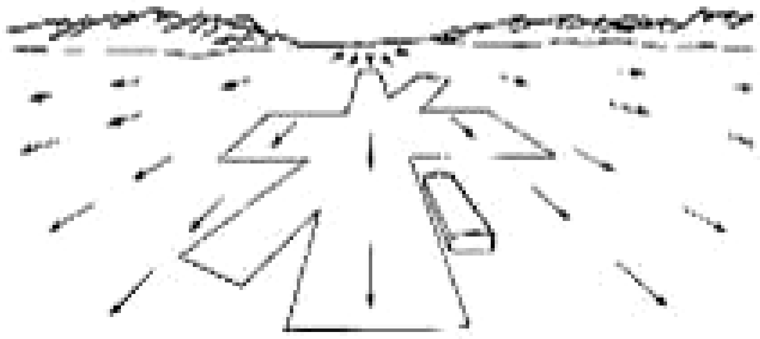
https://www.cnblogs.com/SCCQ/p/12305162.html

光流实例

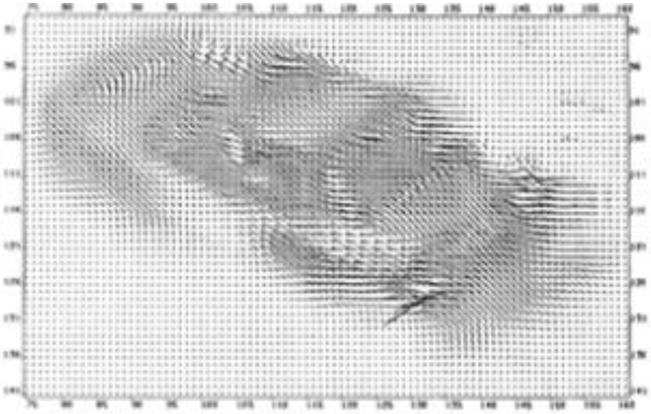
from S. Seitz, R. Szeliski, M. Pollefeys

计算光流的目的

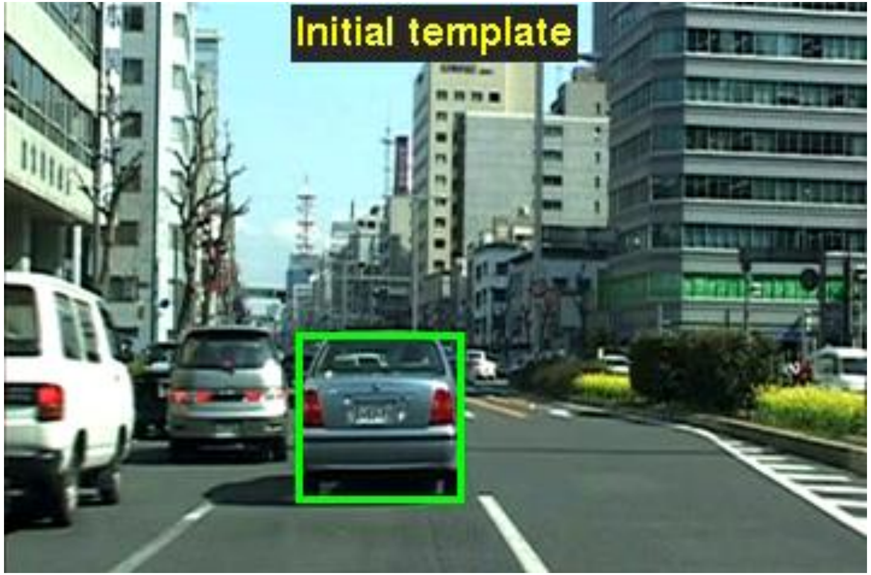
- ◆ 利用光流估计结果求取图像序列中点的对应关系
- ◆ 利用光流估计结果求取相邻帧之间物体的运动信息



2. 光流估计的用途—用于稠密运动估计

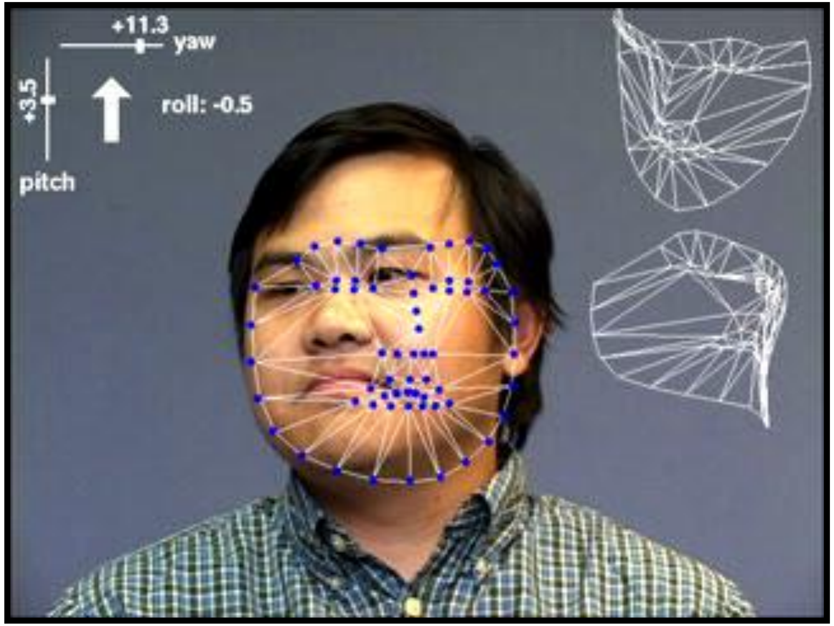


光流估计的用途—刚体运动跟踪



(Comaniciu et al, Siemens)

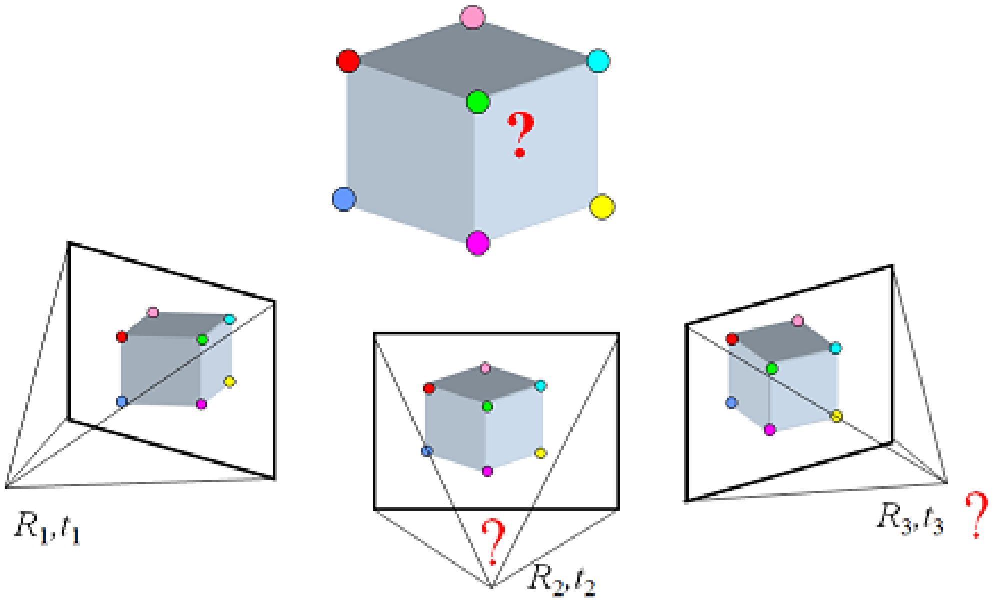
光流估计的用途—人脸跟踪



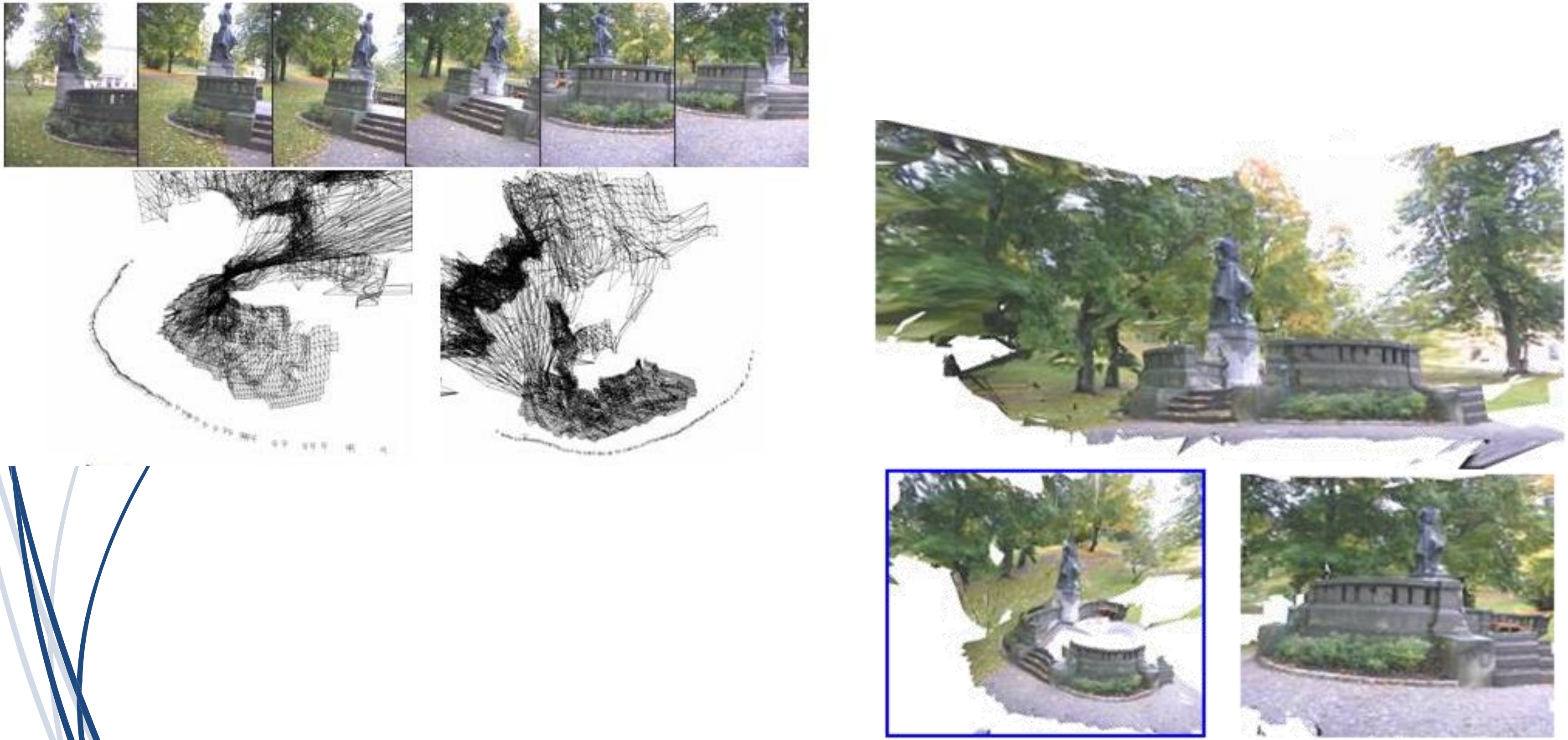
(Simon Baker et al, CMU)

应用于SFM(Structure from Motion)

- 给定两个或多个图像中的一组对应点，计算摄像机参数和三维点坐标

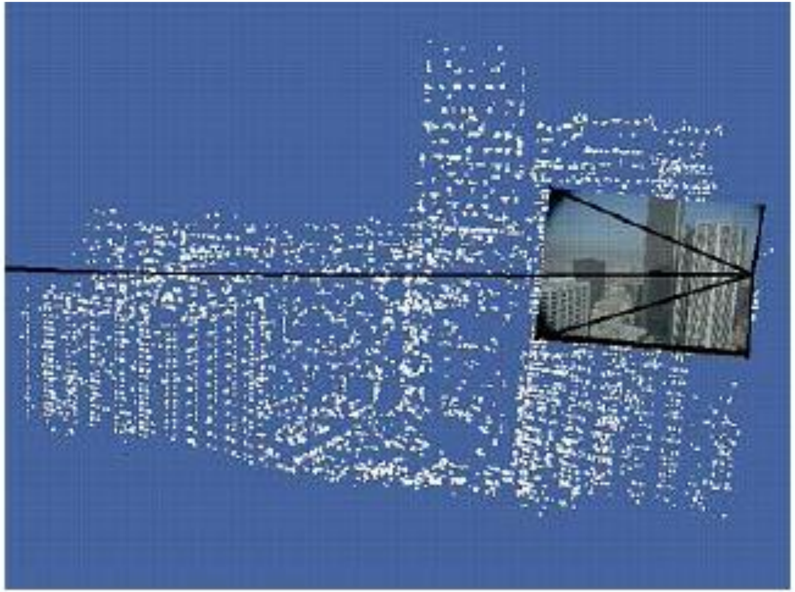


应用于SFM



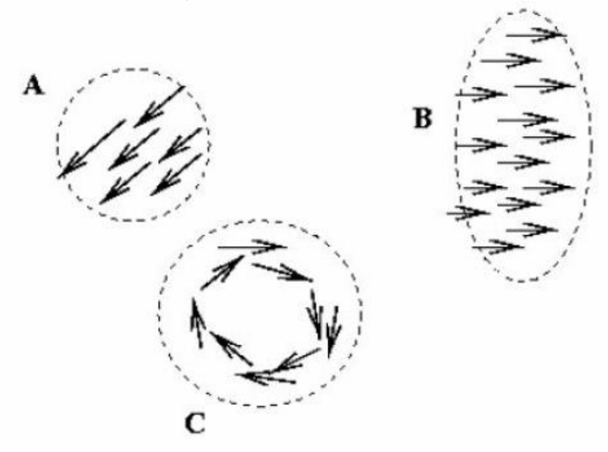
(David Nister, Kentucky)

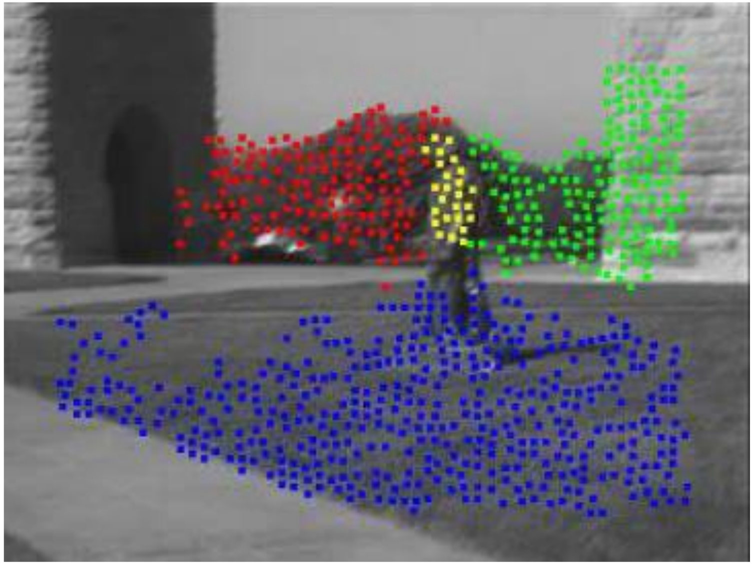
估计场景三维结构



Fei Fei-Li Standford Vision Lab

基于运动的分割





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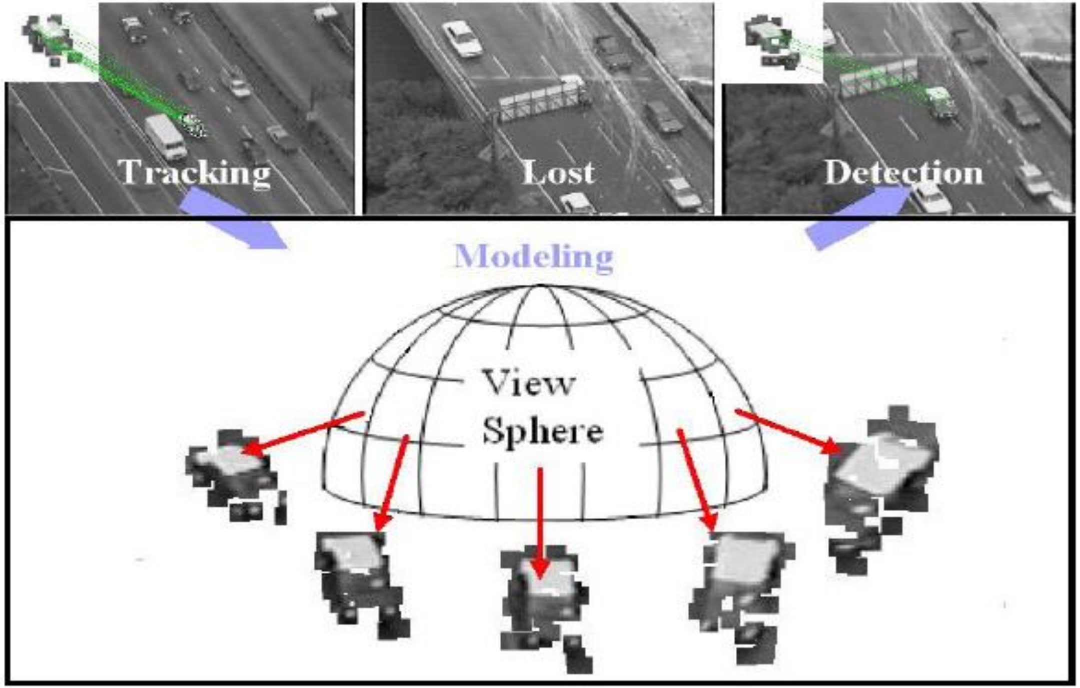
跟踪目标



Z.Yin and R.Collins, "On-the-fly Object Modeling while Tracking," *IEEE Computer Vision and Pattern Recognition (CVPR '07)*, Minneapolis, MN, June 2007.

超分辨率复原

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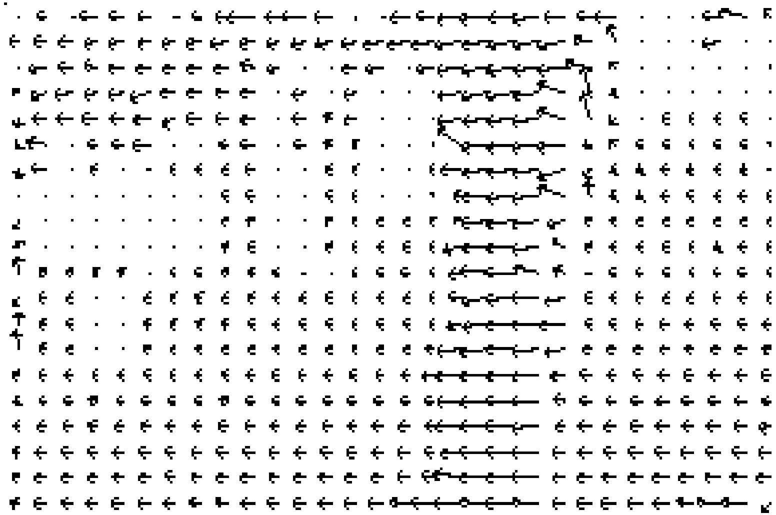
3. 运动场与光流的关系

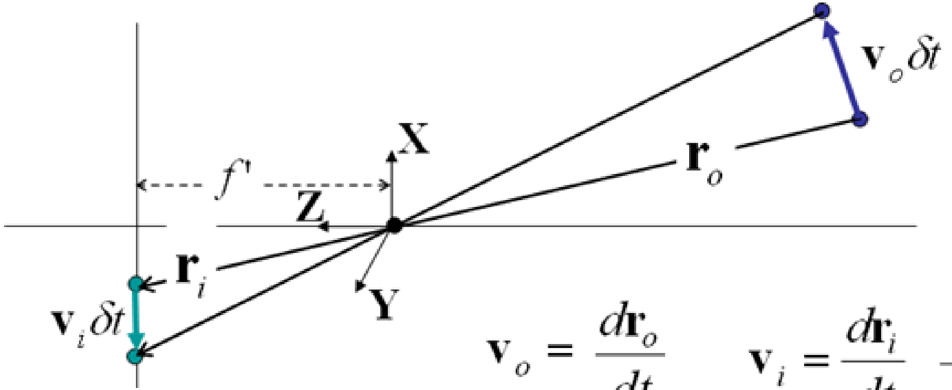
- 光流法是利用图像序列中像素在相邻帧间的变化来找到相邻帧之间点的相关性，从而计算出相邻帧之间物体的运动信息

- 光流的产生：光流是由于场景中前景目标本身的移动、相机的运动，或者两者的共同作用产生的。

光流描述

- 通常将光流视为一个描述该点瞬时速度的二维矢量u=(u, v)，称为光流矢量。
- 一般情况下，2D成像平面大量点构成的速度场，也称为光流场





$$\mathbf{v}_o = \frac{d\mathbf{r}_o}{dt} \qquad \mathbf{v}_i = \frac{d\mathbf{r}_i}{dt} \qquad \frac{1}{f'}\mathbf{r}_i = \frac{\mathbf{r}_o}{\mathbf{r}_o \cdot \mathbf{Z}}$$

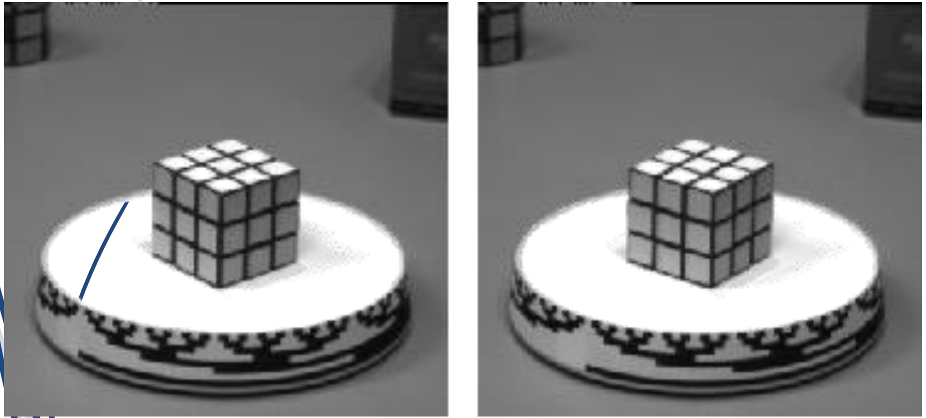
Motion field

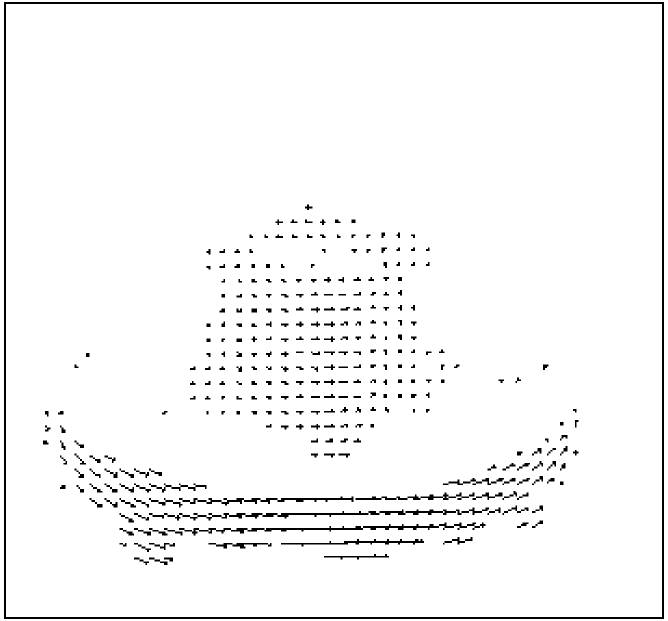
$$\mathbf{v}_i = \frac{d\mathbf{r}_i}{dt} = f'\frac{(\mathbf{r}_o \cdot \mathbf{Z})\mathbf{v}_o - (\mathbf{v}_o \cdot \mathbf{Z})\mathbf{r}_o}{(\mathbf{r}_o \cdot \mathbf{Z})^2} = f'\frac{(\mathbf{r}_o \times \mathbf{v}_o) \times \mathbf{Z}}{(\mathbf{r}_o \cdot \mathbf{Z})^2}$$

光流场与速度场

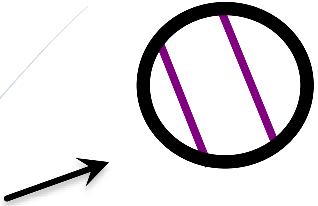
- 理想情况下：运动场与光流场相同

Optical flow = Motion field

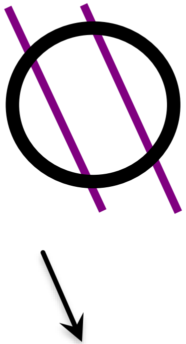




光流估计中的孔径问题(Aperture problem)



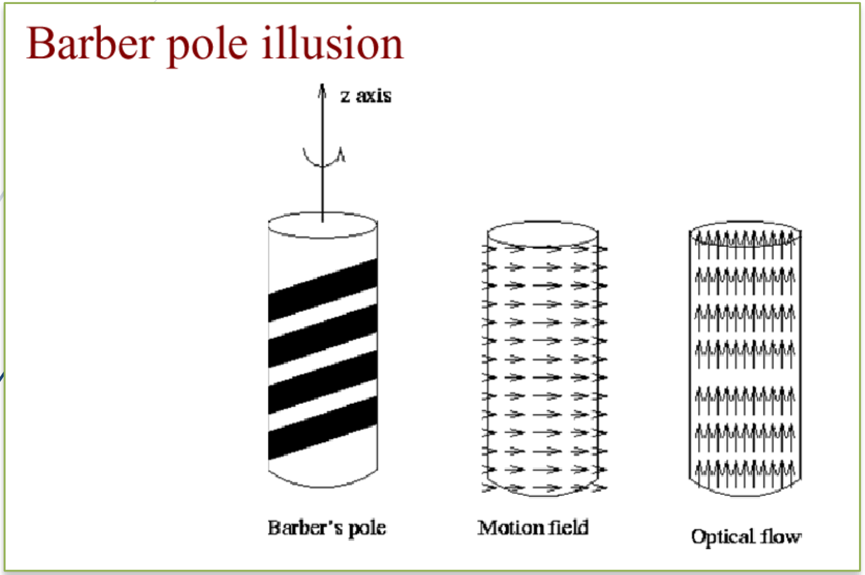
实际运动方向，能感知运动及光流

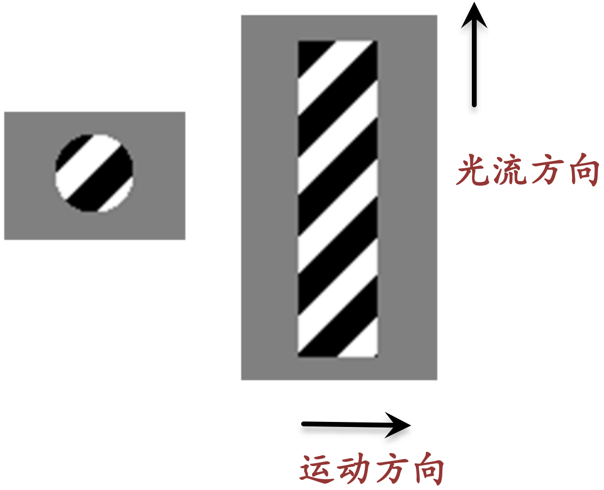


实际运动方向，感知不到运动及光流

- 根本原因：
  - 有运动，感知不到光流
  - 需要足够的图像梯度信息，才能进行准确的估计

理发杆幻觉现象（The barber pole illusion）





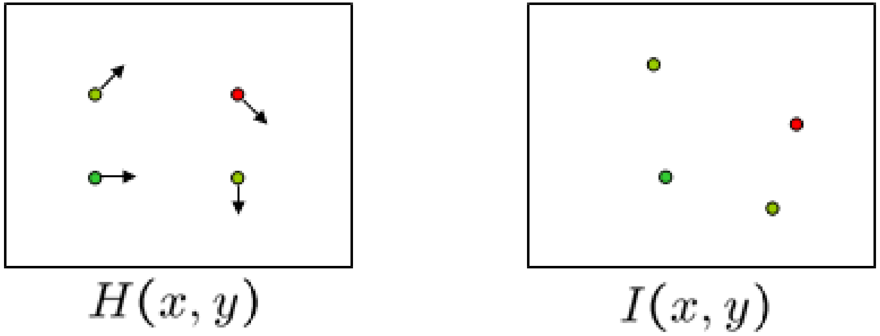
http://en.wikipedia.org/wiki/Barberpole_illusion

4. 光流的估计方法

- 给定两个相继帧，估计2D场景任意一点P(x, y)的水平和垂直速度（u, v）

  其中u和v与P的位置有关，我们记为u（x， y）和v（x， y）

- 光流估计目标： 计算出u和v

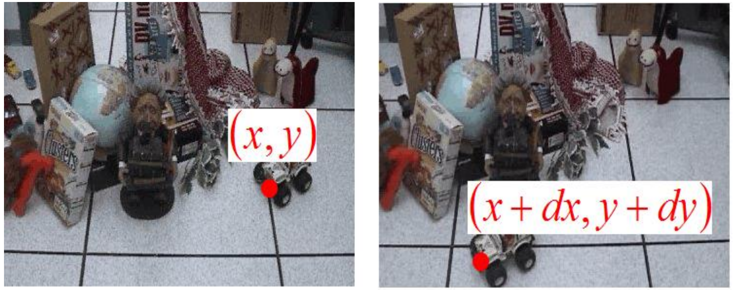


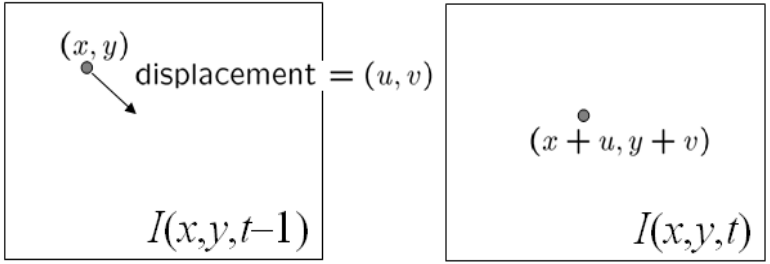
（1）光流估计的三个重要假设

假设计算光流的点我们记为P：

- 亮度恒定： P点在不同帧中亮度不变
- 小动作： P点的运动位移很小
- 相干性： P点运动与其邻域像素的运动具有相似性

亮度恒定的假设：





- 亮度一致性方程

$$I(x, y, t-1) = I(x+u(x, y), y+v(x, y), t)$$

（2）光流估计约束

$$I(x,y,t-1)=I(x+u(x,y),y+v(x,y),t)$$

◆ 右侧线性化，用泰勒展开

$$I(x,y,t-1)\approx I(x,y,t)+I_x u(x,y)+I_y v(x,y)$$

◆ 亮度恒定，得到： $I_x u+I_y v+I_t\approx 0$

即 $\nabla I\cdot(u,v)+I_t=0$

I_x, I_y, I_t 利用梯度计算

I_t图像上某点P在时间序列上的灰度变化

(u, v) 表示某点P的运动速度

（ I_x, I_y）表示某点P的灰度在水平和垂直方向的变化（水平差分和垂直差分）

光流估计

- **光流估计目标： 计算出u和v**
- 出现的问题：图像中的每一点上有两个未知数u 和v ，但只有一个方程，因此，只是利用一个像素点信息不能确定光流（u, v）.
  - 最优化问题，迭代求解
  - 增加约束，使问题可解

（3）光流估计

- 光流约束中的误差公式：

$$e_c = \iint_{image} (I_x u + I_y v + I_t)^2 \, dx \, dy$$

- 需要附加额外约束

- 平滑项约束：

$$e_s = \iint_{image} (u_x^2 + u_y^2) + (v_x^2 + v_y^2) \ \ dx \, dy$$

光流估计常用方法

- 基于梯度的方法
  - Horn-Schunck法
  - Lucas-Kanade方法
  - Nagel方法
- 块匹配方法

（a）Horn-Schunck法计算光流-基于梯度方法

- 根据光流约束方程

$$e^2(\mathbf{x}) = \left(I_x u + I_y v + I_t\right)^2 \qquad \mathbf{x} = (x, y)^T$$

- 根据光滑变化的假设

$$s^2(\mathbf{x}) = \iint\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2\right]dxdy$$

- 能量项：将光滑性与微分约束加权组合

$$E = \iint\left\{e^2(\mathbf{x}) + \alpha s^2(\mathbf{x})\right\}dxdy$$

- 变分法迭代求解方法：

$$u^{n+1} = \bar{u}^n - I_x \frac{I_x \bar{u}^n + I_y \bar{v}^n + I_t}{\alpha + I_x^2 + I_y^2}$$

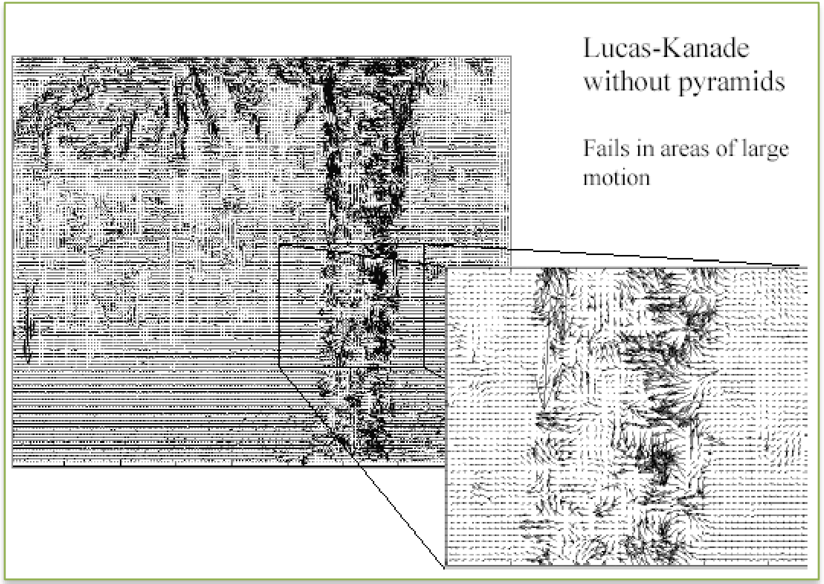
$$v^{n+1} = \bar{v}^n - I_y \frac{I_x \bar{u}^n + I_y \bar{v}^n + I_t}{\alpha + I_x^2 + I_y^2}$$

Lucas Kanade方法—差分方法

- 1981年 由Bruce D. Lucas和Takeo Kanade发明
- 主要思想：假设在一个小的空间邻域内运动矢量保持恒定，即某点P与临像素具有相同的运动
- 使用加权最小二乘方法估计光流。在一个小的空间邻域上，对光流估计误差进行优化处理：

$$\sum_{(x,y)\in\Omega} W^2(\mathbf{x})(I_x u + I_y v + I_t)^2$$

其中$W(\mathbf{x})$表示窗口权重函数



Lucas Kanade方法存在的问题

- 运动小速度，亮度不变、区域一致性均不容易满足
- 如当物体运动速度较快时，假设不成立，那么就产生较大的偏差，为了克服这一问题，减少图像中物体的运动速度，采取措施：缩小图像的尺寸。
- 改进措施：生成金字塔图像，逐层求解，不断精确来求得。

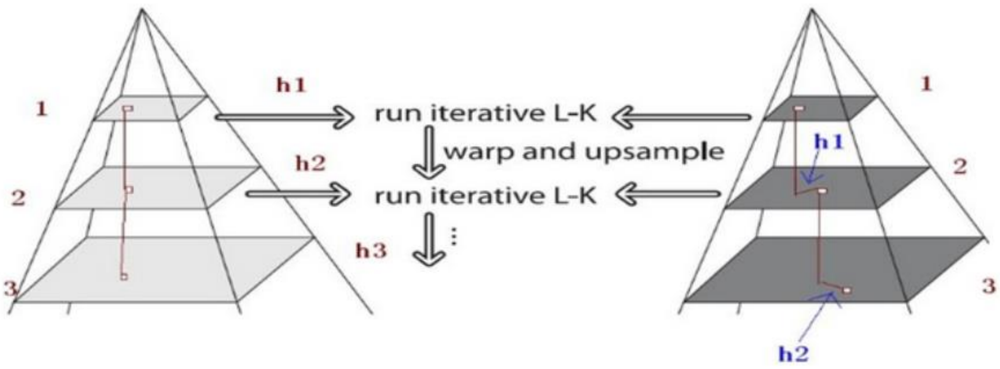
Lucas-Kanade改进算法

- 主要的步骤有三步：
  - 建立金字塔
  - 金字塔中对求解跟踪
  - 迭代过程

建立金字塔

- 用一个[0.25 0.5 0.25]的低通滤波器对I_{L-1}进行卷积

$$I^L = \frac{1}{4} I^{L-1}(2x, 2y) +$$
$$\frac{1}{8}\left(I^{L-1}(2x-1, 2y) + I^{L-1}(2x+1, 2y) + I^{L-1}(2x, 2y-1) + I^{L-1}(2x, 2y+1)\right) +$$
$$\frac{1}{16}\left(I^{L-1}(2x-1, 2y-1) + I^{L-1}(2x+1, 2y-1) + I^{L-1}(2x-1, 2y+1) + I^{L-1}(2x+1, 2y+1)\right)$$

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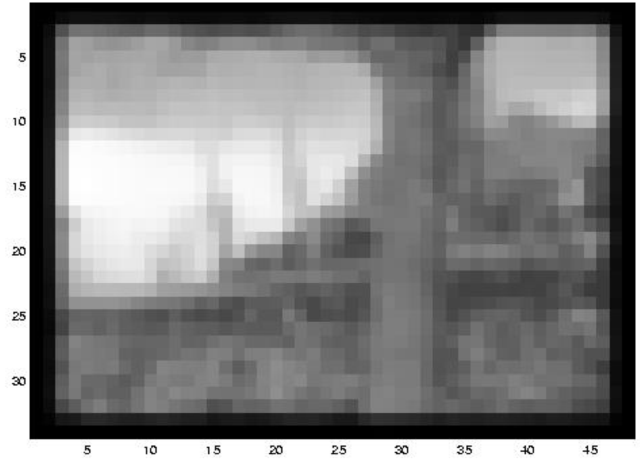
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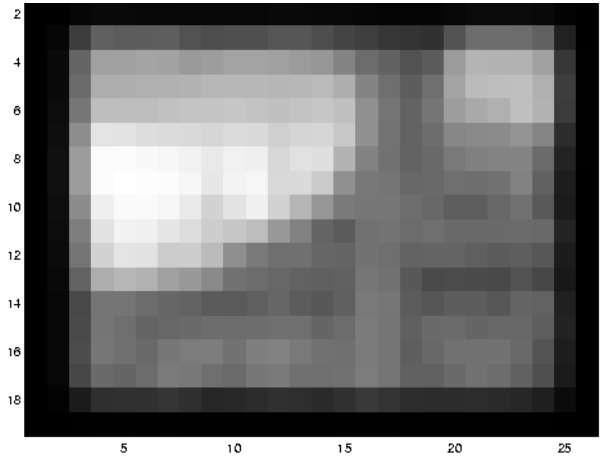
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金字塔跟踪

- 首先，光流和仿射变换矩阵在最高一层的图像上计算出；
- 将上一层的计算结果传递给当前层作为初始值，进一步在这个初始值的基础上，计算当前层的光流和仿射变化矩阵；
- 再将当前层的光流和仿射矩阵作为初始值传递给下一层图像，
- 直到传递给最后一层，即原始图像层，计算出来的光流和仿射变换矩阵作为最后的光流和仿射变换矩阵的结果。

迭代过程

- 在金字塔的每一层，计算出光流dL和仿射变换矩阵AL，目标使误差ξL最小

- 方法： 将上一层的光流u传给当前层，计算当前层像素强度,计算出图像在该点x方向和y方向上的偏导

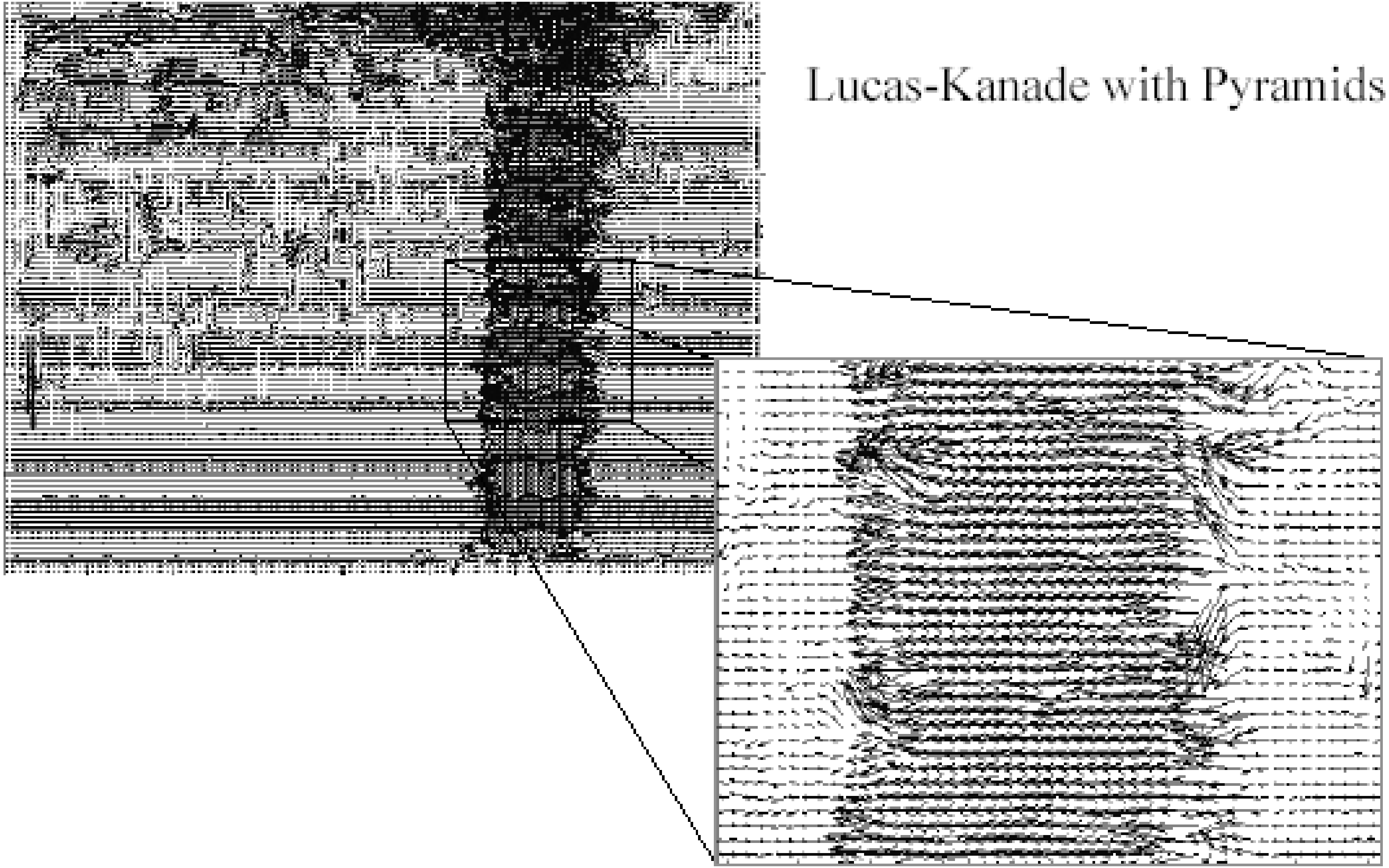
$$I(x) \leftarrow I^L(x + u / 2^L)$$

Ix=[I(x+1,y)-I(x-1,y)]/2

Iy=[I(x,y+1)-I(x,y-1)]/2

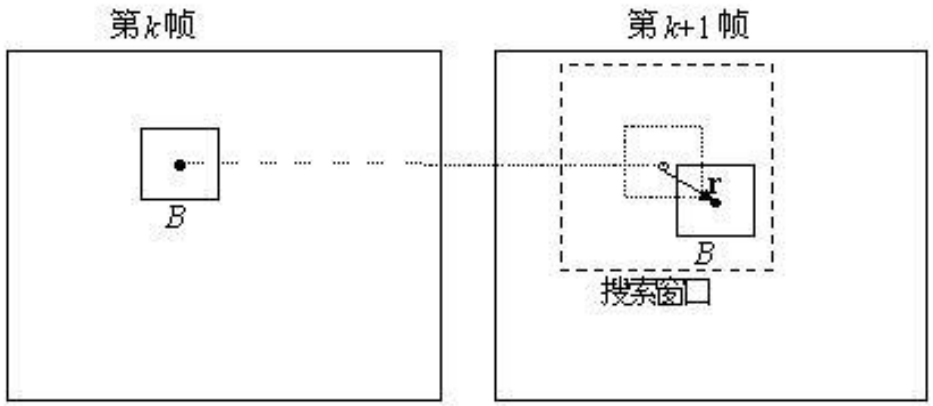
- 根据两帧图像间相同位置点的灰度值之差，更新跟踪结果，不断迭代

Lucas-Kanade with Pyramids



（b）块匹配方法

- 将图像帧划分为块
- 对于当前帧中的图像块，搜索其在下一帧中的对应块
- 计算对应块位移矢量



- 块匹配算法的几个问题：
  - 匹配准则
  - 搜索策略
  - 块尺寸选择方法

块匹配准则

1. 最小均方差（MSE）

$$MSE(\Delta x, \Delta y) = \frac{1}{mn} \sum_{(x,y)\in W} [I(x, y, k) - I(x + \Delta x, y + \Delta y, k + 1)]^2$$

2. 最小平均绝对差（MAD）

$$MAD(\Delta x, \Delta y) = \frac{1}{mn} \sum_{(x,y)\in W} | I(x, y, k) - I(x + \Delta x, y + \Delta y, k + 1) |$$

3. 最大匹配像素（MPC）

$$MPC(\Delta x, \Delta y) = \sum_{(x,y)\in W} T(x, y, \Delta x, \Delta y) \qquad [\Delta x, \Delta y]^T = \arg \min_{(\Delta x, Xy)} MPC(\Delta x, \Delta y)$$

$$T(x, y, \Delta x, \Delta y) = \begin{cases} 1 & \text{如果} \ | I(x, y, k) - I(x + \Delta x, y + \Delta y, k + 1) | \le T \\ 0 & \text{其它} \end{cases}$$

搜索策略

- 全搜索策略：在图像全局范围计算所有可能的位移矢量对应的匹配误差，然后选择最小匹配误差对应的矢量就是最佳位移估计值

- 优点：可以找到全局最优值，但十分浪费时间，实际中经常采用快速搜索策略

5. 密集光流（稠密光流）与稀疏光流

- **稠密光流：**对图像进行逐点求取光流，实现逐点匹配，达到图像配准目的。
- **稀疏光流：** 针对图像上若干个特征点。
- **区别：** 稠密光流计算图像上所有点，形成一个稠密的光流场，其配准后的效果也明显优于稀疏光流配准的效果。但是存在计算量大的问题。

从稀疏光流到稠密光流

- 局部加权线性回归插值稠密光流场
- 稀疏特征点聚类的流向量

OpenCV计算光流的函数

1）calcOpticalFlowPyrLK ：通过金字塔Lucas-Kanade 光流方法计算某些点集的光流（稀疏光流）。

2）calcOpticalFlowFarneback：用Gunnar Farneback 的算法计算稠密光流（即图像上所有像素点的光流都计算出来）

3）CalcOpticalFlowBM：通过块匹配的方法来计算光流。

4）CalcOpticalFlowHS：用Horn-Schunck算法计算稠密光流。

5）calcOpticalFlowSF： 非递归的优化方法。

6. 智能光流估计技术及应用

(1)FlowNet

- 基于深度学习的光流估计算法FlowNet，收录于ICCV2015。
- 将光流估计问题作为监督学习任务来解决的CNN。
- 提出并比较了两种架构，一种通用架构，另一种包括在不同图像位置关联特征向量的层。
- 生成了一个大型合成 Flying Chairs数据集。

FlowNet: Learning Optical Flow with Convolutional Networks

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Abstract

Convolutional neural networks (CNNs) have recently been very successful in a variety of computer vision tasks, especially on those linked to recognition. Optical flow estimation has not been among the tasks CNNs succeeded at. In this paper we construct CNNs which are capable of solving the optical flow estimation problem as a supervised learning task. We propose and compare two architectures: a generic architecture and another one including a layer that correlates feature vectors at different image locations. Since existing ground truth data sets are not sufficiently large to train a CNN, we generate a large synthetic Flying Chairs dataset. We show that networks trained on this unrealistic data still generalize very well to existing datasets such as Sintel and KITTI, achieving competitive accuracy at frame rates of 5 to 10 fps.

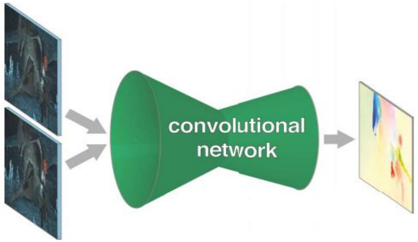


Figure 1. We present neural networks which learn to estimate optical flow, being trained end-to-end. The information is first spatially compressed in a contractive part of the network and then refined in an expanding part.

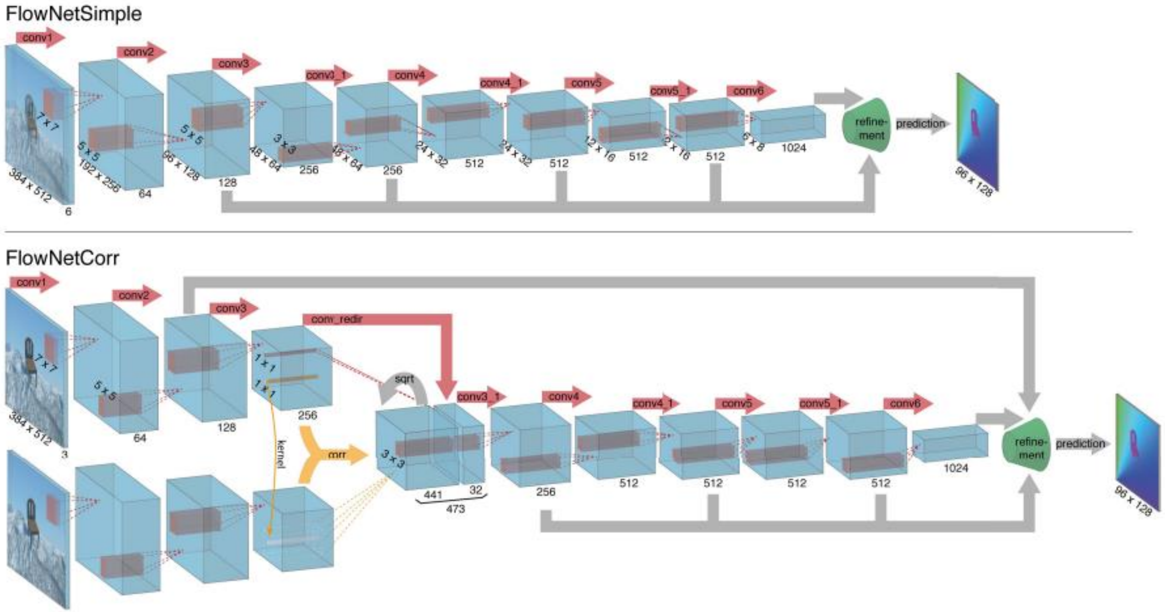
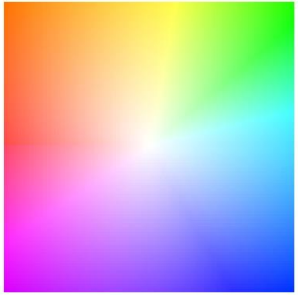


Figure 2. The two network architectures: FlowNetSimple (top) and FlowNetCorr (bottom). The green funnel is a placeholder for the expanding refinement part shown in Fig 3. The networks including the refinement part are trained end-to-end.

Figure 5. **Flying Chairs.** Generated image pair and color coded flow field (first three columns), augmented image pair and corresponding color coded flow field respectively (last three columns).



(2)FlowNet2.0

- 进一步提升了算法的速度和光流质量。
- FlowNet 2.0增加了训练数据，并使用更复杂的训练策略提升性能。
- 针对小位移的情况引入特定子网络进行处理。
- FlowNet 2.0牺牲运行速度，提高准确性
- 各个公开数据集上，效果较好。

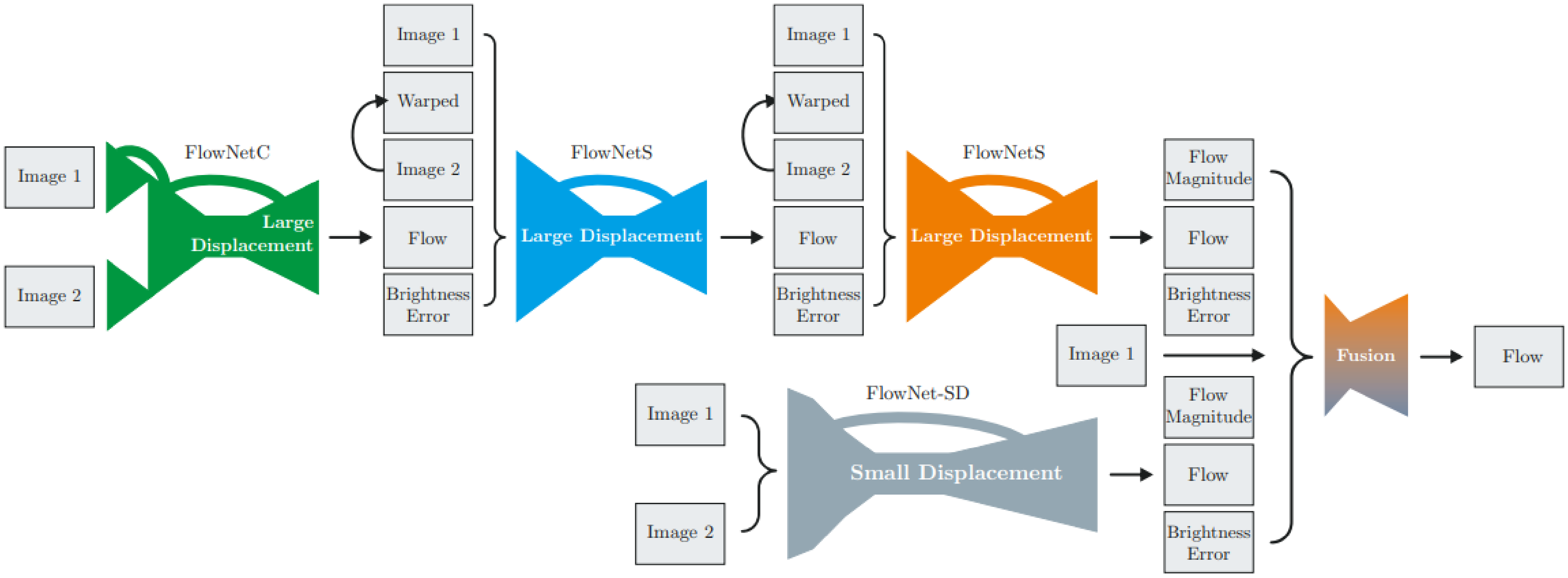


Figure 2. Schematic view of complete architecture: To compute large displacement optical flow we combine multiple FlowNets. Braces indicate concatenation of inputs. *Brightness Error* is the difference between the first image and the second image warped with the previously estimated flow. To optimally deal with small displacements, we introduce smaller strides in the beginning and convolutions between upconvolutions into the FlowNetS architecture. Finally we apply a small fusion network to provide the final estimate.

(3) 光流在人脸表情分析中的应用

Key words:
expression, motion

SelfME: Self-Supervised Motion Learning for Micro-Expression Recognition

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Abstract

Facial micro-expressions (MEs) refer to brief spontaneous facial movements that can reveal a person's genuine emotion. They are valuable in lie detection, criminal analysis, and other areas. While deep learning-based ME recognition (MER) methods achieved impressive success, these methods typically require pre-processing using conventional optical flow-based methods to extract facial motions as inputs. To overcome this limitation, we proposed a novel MER framework using self-supervised learn-

Figure 1. MEs may be imperceptible to the naked eye, but the motion between the onset (the moment when the facial action begins to grow stronger) and the apex (the moment when the facial action reaches its maximum intensity) makes them readily observable. SelfME learns this motion automatically.

光流在人脸表情分析中的应用

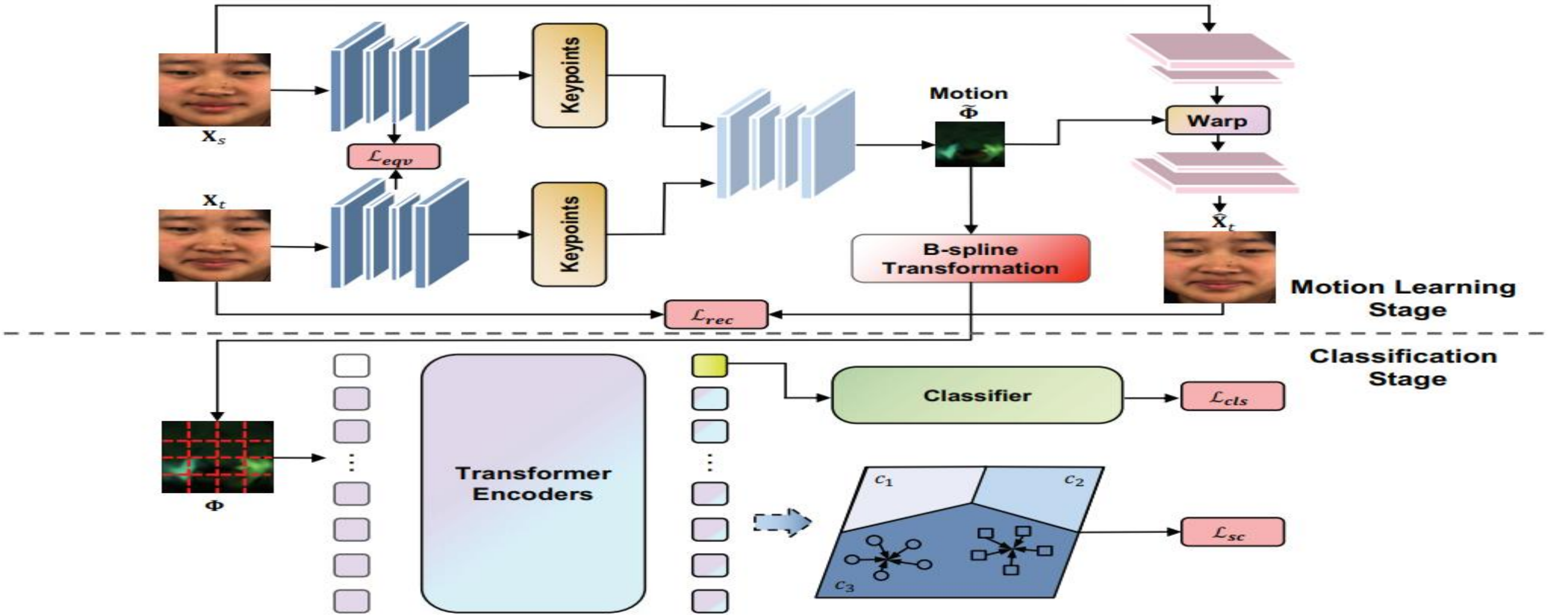

Figure 2. Framework of SelfME. SelfME consists of two stages: a motion learning stage and a classification stage. In the **motion learning stage**, the self-supervised motion learner was trained on a source frame $\mathbf{X}_s$ and a target frame $\mathbf{X}_t$ in ME sequences with a reconstruction objective, to warp the source frame $\mathbf{X}_s$ into the estimated target frame $\hat{\mathbf{X}}_t$. A B-spline transformation expands the size of the motion field to a similar size as the original image input, producing the improved motions Φ that are fed into the classification stage. In the **classification stage**, an SCViT was used to constrain the learning of similar facial action features for the left and right parts of faces.

微表情识别中应用

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Micro and Macro Facial Expression Recognition Using Advanced Local Motion Patterns

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Abstract—In this paper, we develop a new method that recognizes facial expressions, on the basis of an innovative Local Motion Patterns (LMP) feature. The LMP feature analyzes locally the motion distribution in order to separate consistent mouvement patterns from noise. Indeed, facial motion extracted from the face is generally noisy and without specific processing, it can hardly cope with expression recognition requirements especially for micro-expressions. Direction and magnitude statistical profiles are jointly analyzed in order to filter out noise. This work presents three main contributions. The first one is the analysis of the face skin temporal elasticity and face deformations during expression. The second one is a unified approach for both macro and micro expression recognition leading the way to supporting a wide range of expression intensities. The third one is the step forward towards in-the-wild expression recognition, dealing with challenges such as various intensity and various expression activation patterns, illumination variations and small head pose variations. Our method outperforms state-of-the-art methods for micro expression recognition and positions itself among top-ranked state-of-the-art methods for macro expression recognition.

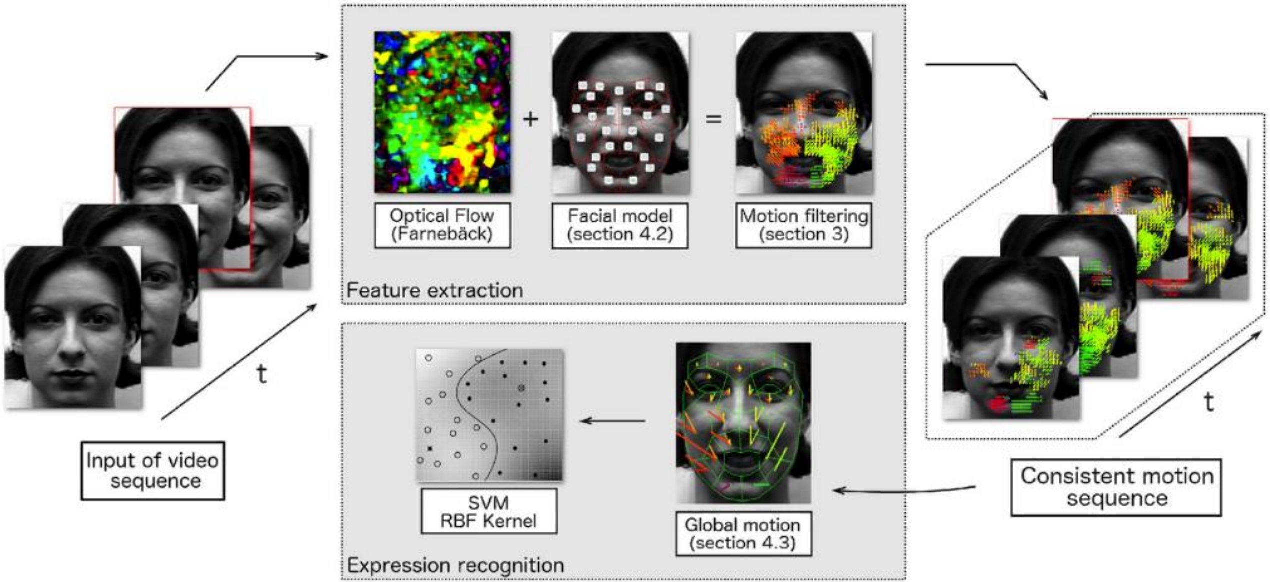

Fig. 1. Overview of our expression recognition method.